

Every receipt, sealed.

Proof of purchase becomes proof of ownership.

A receipt photo proves nothing.



50.1%

of people can spot the fake. A coin flip.

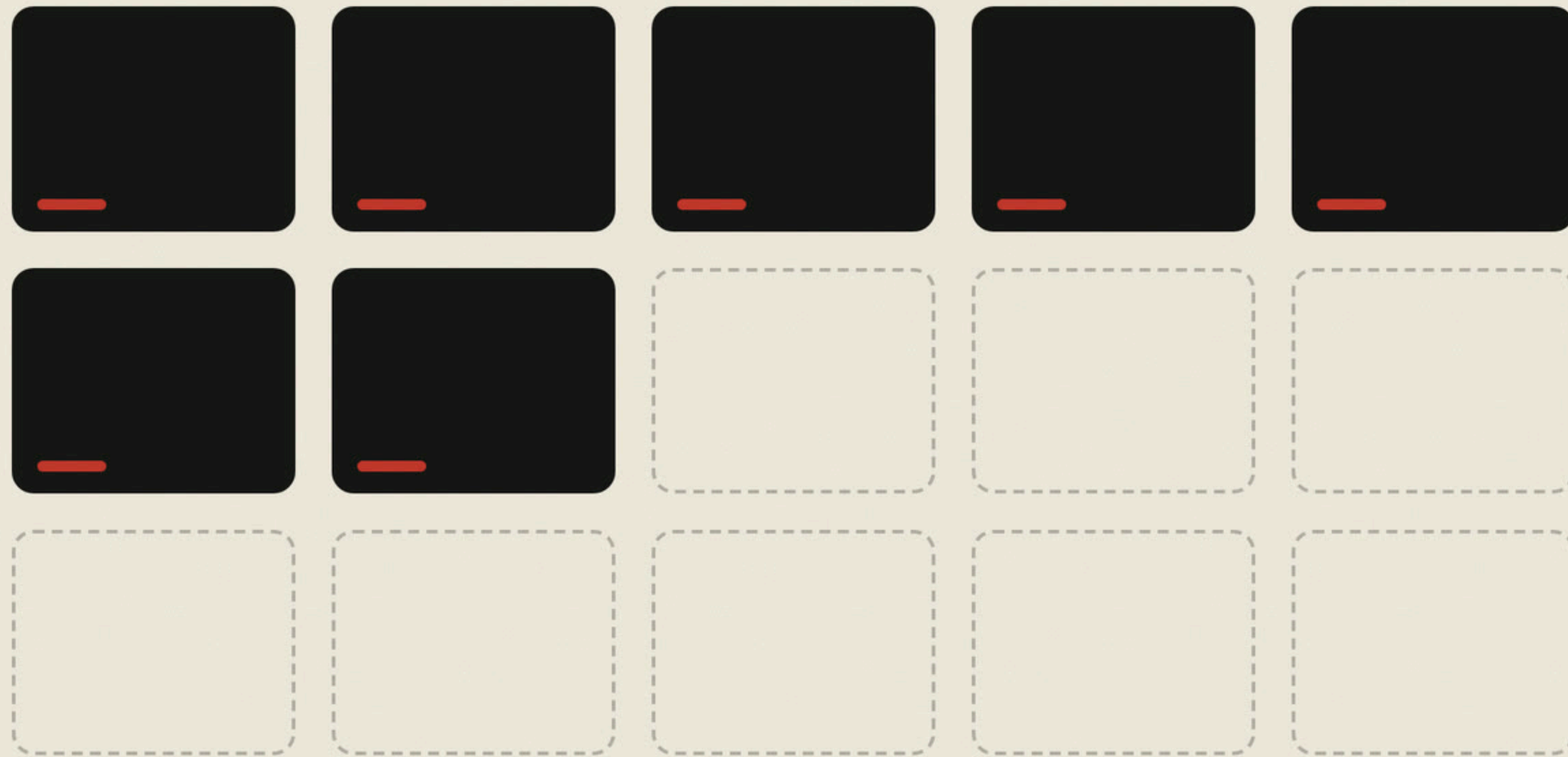
+244%

digital document forgeries, year on year

0.53–0.60

best forensic detectors (AUC). Barely above chance.

Points never feel like yours.



14.8 loyalty programs joined. **6.7** actually used. The rest sit forgotten.

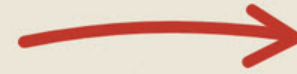
73%

find loyalty programs too complicated

~60%

of coalition programs fail within ten years

Seal the receipt. Own the brand.



+15 sbNKE · vault shares

読

Scan

The merchant signs the receipt

証

Prove

Checked once, on Arbitrum

有

Own

Brand shares in your wallet

Four guarantees. One transaction.



Signature, not pixels

Change one digit and the claim fails.

```
keccak256(  
  tag,  
  merchant,  
  receipt)
```

Counted once, for anyone

The receipt's nullifier burns on-chain.



Yours. No admin.

Admin-less ERC-4626 vault shares.



9.2x cheaper to verify

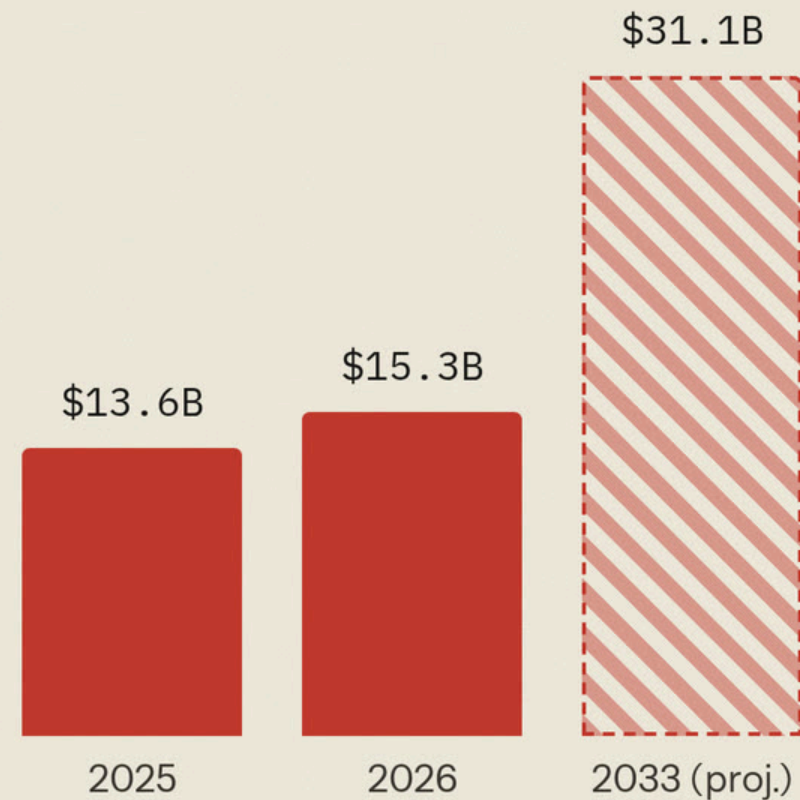
Ed25519 in Rust on Arbitrum Stylus.

The rails already exist.

24.07B

UPI payments in India in September 2026 alone. Each one is a purchase that could be sealed.

NPCI UPI product statistics



Loyalty software market, growing 10.7% a year

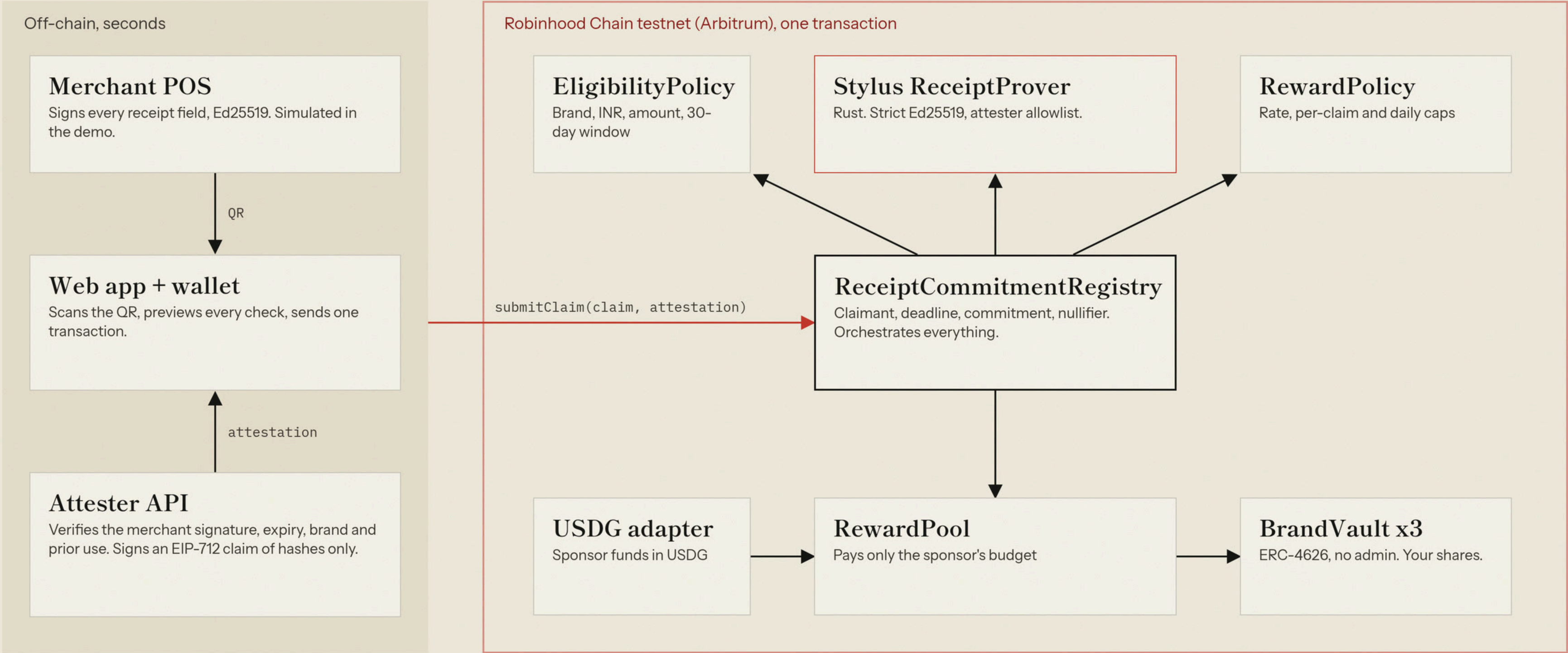
Grand View Research, 2026

+244%

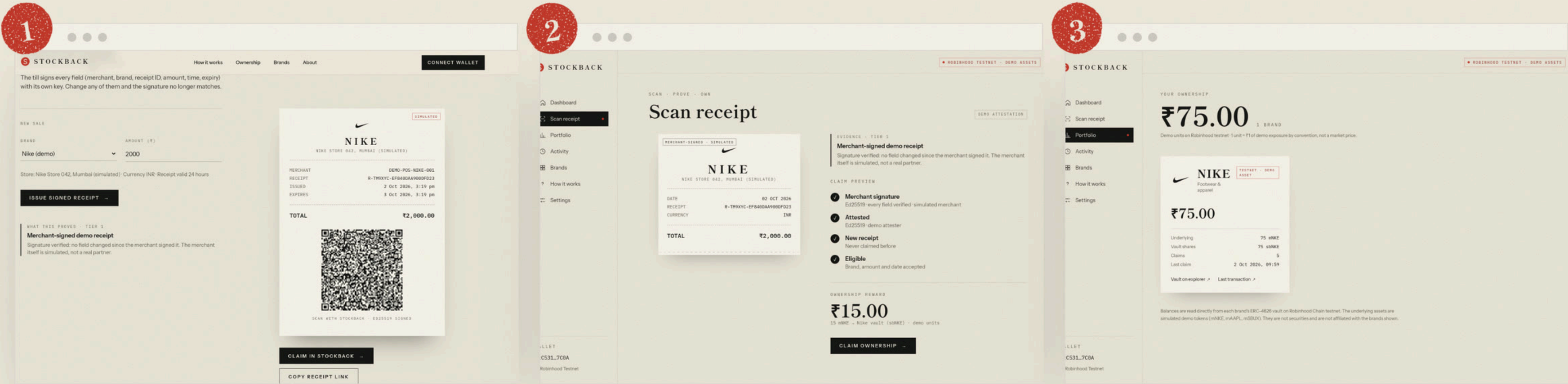
digital document forgeries, year on year. Verified receipts become a need.

Entrust 2025, via arXiv:2604.25213

Architecture



The workflow, live.



The till prints a signed QR

`stockbacks.vercel.app/merchant`

Signature verified

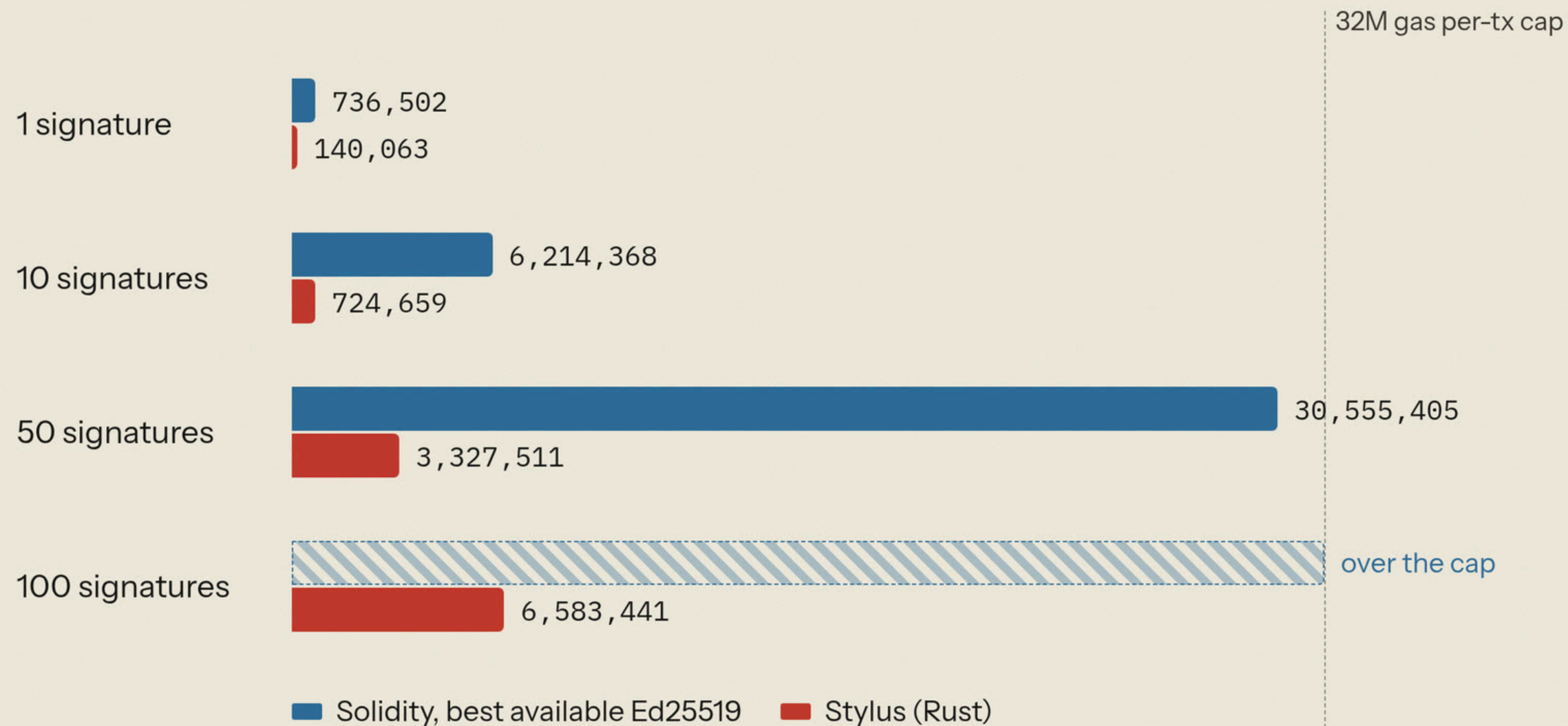
`stockbacks.vercel.app/app/scan`

Vault shares in your wallet

`stockbacks.vercel.app/app/portfolio`

About 30 seconds end to end. Tampered or replayed QR codes are rejected before anything is signed.

Built, deployed, measured.



9.2x
less gas than Solidity

130
automated tests: 83 Solidity, 5 Rust, 42 web

9
contracts: 8 Solidity, 1 Rust on Stylus

Everyone keeps what's theirs.

Shoppers

- ✓ A share of the brand, not a point
- ✓ No expiry, no admin can dilute it
- ✓ No personal data on-chain
- ✓ One wallet for every brand



Brands

- ✓ Reward only signed, verified purchases
- ✓ Budgets and daily caps they set
- ✓ No coalition operator taking a cut
- ✓ Fund rewards in USDG



Where it goes next.



Now

Live on testnet

Merchant-signed receipts, Stylus verifier, admin-less vaults, USDG adapter



Next

Production-ready

Hold period with refund voiding, real POS keys, k-of-n attestors



Later

At scale

zkTLS payment proofs, mainnet with issuer-authorized assets



Scan. Prove. Own.

stockbacks.vercel.app github.com/notwen123/STOCKBACK

